

Congruence Classes and Solving Linear Congruences

Lecture based on Sections 1.4 and 1.5 of *Numbers, Groups and Codes*

Applied Algebra
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1 Congruence Classes

2 Solving Linear Congruences

- 1 Congruence modulo n and congruence classes
- 2 Arithmetic in $\mathbb{Z}_n$
- 3 Invertible classes and zero-divisors
- 4 Linear congruences $ax \equiv b \pmod{n}$
- 5 Chinese Remainder Theorem and worked examples

Learning goals

By the end of the lecture, students should be able to:

- ▶ define $a \equiv b \pmod{n}$ and describe congruence classes,
- ▶ compute in $\mathbb{Z}_n$ using class arithmetic,
- ▶ decide when a class has a multiplicative inverse,
- ▶ distinguish invertible classes from zero-divisors,
- ▶ solve $ax \equiv b \pmod{n}$ in all cases,
- ▶ solve compatible systems of congruences with coprime moduli.

Prerequisites to recall

We will use the following facts from earlier sections.

Facts in the background

- ▶ $a \mid b$ means: there exists $k \in \mathbb{Z}$ such that $b = ak$.
- ▶ $d = \gcd(a, n)$ is the smallest positive integral linear combination of a and n .
- ▶ If $\gcd(a, n) = 1$, then there exist integers r, s with $ar + ns = 1$.
- ▶ The Euclidean algorithm gives both $\gcd(a, n)$ and a Bézout identity.

Why this matters

That last identity is exactly what produces inverses modulo n .

Why congruence classes?

- ▶ Clock arithmetic: $17 \equiv 5 \pmod{12}$.
- ▶ Last digits: 37^2 and 7^2 have the same last digit.
- ▶ Remainders often matter more than exact size.
- ▶ Number theory becomes cleaner when we group integers by common remainder.

Key viewpoint

Instead of working with all integers separately, we work with **classes of integers that leave the same remainder modulo n** .

Definition of congruence modulo n

Let $n \geq 2$ be fixed.

Definition

For integers a, b , we say

$$a \equiv b \pmod{n}$$

if and only if $n \mid (a - b)$.

Equivalent phrasing:

- ▶ a and b have the same remainder on division by n ;
- ▶ $a - b$ is a multiple of n ;
- ▶ $a = b + kn$ for some $k \in \mathbb{Z}$.

First examples of congruence

$$17 \equiv 5 \pmod{12}$$

$$\text{since } 17 - 5 = 12,$$

$$31 \equiv 3 \pmod{7}$$

$$\text{since } 31 - 3 = 28 = 4 \cdot 7,$$

$$-2 \equiv 8 \pmod{5}$$

$$\text{since } -2 - 8 = -10 = -2 \cdot 5,$$

$$30 \equiv 6 \pmod{8}$$

$$\text{since } 30 - 6 = 24 = 3 \cdot 8.$$

Important

Congruence is about *difference*, not equality.

For example, $17 \neq 5$ in $\mathbb{Z}$, but $17 \equiv 5 \pmod{12}$.

Congruence as an equivalence relation

For fixed n , the relation $\equiv \pmod{n}$ has three properties:

- ① **Reflexive:** $a \equiv a \pmod{n}$.
- ② **Symmetric:** if $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$.
- ③ **Transitive:** if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.

Consequence

The integers split into disjoint families, called **congruence classes modulo n** .

Congruence classes

Definition

The congruence class of a modulo n is

$$[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod{n}\}.$$

So $[a]_n$ contains all integers with the same remainder as a upon division by n .

Examples

$$[2]_5 = \{\dots, -8, -3, 2, 7, 12, 17, \dots\}$$

$$[-1]_6 = [5]_6 = \{\dots, -7, -1, 5, 11, 17, \dots\}$$

All classes modulo 4

There are exactly four congruence classes modulo 4:

$$[0]_4, \quad [1]_4, \quad [2]_4, \quad [3]_4.$$

$$[0]_4 = \{\dots, -8, -4, 0, 4, 8, 12, \dots\},$$

$$[1]_4 = \{\dots, -7, -3, 1, 5, 9, 13, \dots\},$$

$$[2]_4 = \{\dots, -6, -2, 2, 6, 10, 14, \dots\},$$

$$[3]_4 = \{\dots, -5, -1, 3, 7, 11, 15, \dots\}.$$

Observation

Every integer belongs to exactly one of these four classes.

Standard representatives and $\mathbb{Z}_n$

Because there are exactly n classes modulo n , it is convenient to write

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}.$$

Standard representatives

We usually choose $0, 1, \dots, n-1$ as standard representatives.

Examples:

$$[8]_5 = [3]_5, \quad [-7]_5 = [3]_5, \quad [14]_6 = [2]_6.$$

Arithmetic of classes

We define addition and multiplication on classes by

$$[a]_n + [b]_n = [a + b]_n, \quad [a]_n [b]_n = [ab]_n.$$

Question

Do these definitions depend on the chosen representatives?

If $[a]_n = [a']_n$ and $[b]_n = [b']_n$, we need to know that

$$[a + b]_n = [a' + b']_n \quad \text{and} \quad [ab]_n = [a'b']_n.$$

This is the **well-definedness** issue.

Why addition and multiplication are well defined

Suppose

$$a \equiv a' \pmod{n}, \quad b \equiv b' \pmod{n}.$$

Then $n \mid (a - a')$ and $n \mid (b - b')$.

$$\begin{aligned}(a + b) - (a' + b') &= (a - a') + (b - b'), \\ ab - a'b' &= a(b - b') + b'(a - a').\end{aligned}$$

Each right-hand side is divisible by n , so

$$a + b \equiv a' + b' \pmod{n}, \quad ab \equiv a'b' \pmod{n}.$$

Conclusion

Arithmetic in $\mathbb{Z}_n$ does *not* depend on which representatives we pick.

Example: addition in $\mathbb{Z}_5$

$$[4]_5 + [3]_5 = [7]_5 = [2]_5.$$

Also,

$$[-1]_5 + [8]_5 = [7]_5 = [2]_5.$$

Same class, different representatives.

Mental model

Add as usual, then replace by the standard representative in $\{0, 1, 2, 3, 4\}$.

Examples:

$$[2]_5 + [4]_5 = [1]_5, \quad [3]_5 + [3]_5 = [1]_5.$$

Example: multiplication in $\mathbb{Z}_5$

$$[4]_5[3]_5 = [12]_5 = [2]_5.$$

More examples:

$$[2]_5[4]_5 = [8]_5 = [3]_5, \quad [3]_5[3]_5 = [9]_5 = [4]_5.$$

Computational rule

Multiply as integers, then reduce modulo n .

Addition table for $\mathbb{Z}_6$

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

- ▶ Every row is a cyclic shift.
- ▶ Additive inverse of $[a]_6$ is $[-a]_6$.

Multiplication table for $\mathbb{Z}_6$

$\cdot$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Notice that some nonzero classes multiply to 0.

$$[2]_6[3]_6 = [0]_6.$$

Cancellation does not always work

Over the integers, if $ac = bc$ and $c \neq 0$, then $a = b$. In modular arithmetic, this can fail.

Example modulo 8

$$30 \equiv 6 \pmod{8}.$$

Multiplying both sides by 2 gives

$$60 \equiv 12 \pmod{8},$$

which is true. But dividing the original congruence by 2 would suggest

$$15 \equiv 3 \pmod{8},$$

which is false.

Lesson

Cancellation is only valid when the factor being cancelled is **invertible modulo n** .

Invertible congruence classes

Definition

A class $[a]_n$ is **invertible** if there exists $[b]_n$ such that

$$[a]_n[b]_n = [1]_n.$$

In that case, $[b]_n$ is the inverse of $[a]_n$ and is written $[a]_n^{-1}$.

Examples:

- ▶ In $\mathbb{Z}_5$, $[2]_5^{-1} = [3]_5$ because $2 \cdot 3 \equiv 1 \pmod{5}$.
- ▶ In $\mathbb{Z}_8$, $[5]_8^{-1} = [5]_8$ because $25 \equiv 1 \pmod{8}$.

Zero-divisors

Definition

A nonzero class $[a]_n$ is a **zero-divisor** if there exists $[b]_n \neq [0]_n$ such that

$$[a]_n[b]_n = [0]_n.$$

Examples in $\mathbb{Z}_8$:

$$[2]_8[4]_8 = [0]_8, \quad [6]_8[4]_8 = [0]_8.$$

Key distinction

Invertible classes behave like nonzero numbers in ordinary arithmetic.
Zero-divisors are exactly what make cancellation dangerous.

Theorem: inverse iff gcd is 1

Theorem

For $n \geq 2$, the class $[a]_n$ is invertible if and only if

$$\gcd(a, n) = 1.$$

Moreover, if integers r, s satisfy

$$ar + ns = 1,$$

then

$$[a]_n^{-1} = [r]_n.$$

Interpretation

Being invertible modulo n is exactly the same as being relatively prime to n .

Why the theorem is true

If $[a]_n$ has an inverse

Suppose

$$[a]_n[k]_n = [1]_n.$$

Then $ak \equiv 1 \pmod{n}$, so

$$ak - 1 = nt$$

for some $t \in \mathbb{Z}$. Hence

$$ak - nt = 1.$$

So 1 is an integral linear combination of a and n , therefore $\gcd(a, n) = 1$.

If $\gcd(a, n) = 1$

Then Bézout gives

$$ar + ns = 1.$$

Reduce modulo n :

$$ar \equiv 1 \pmod{n}.$$

So

$$[a]_n[r]_n = [1]_n,$$

and $[r]_n$ is the inverse.

Worked inverse example: $[6]_{17}$

We want $[6]_{17}^{-1}$.

$$17 = 2 \cdot 6 + 5,$$

$$6 = 1 \cdot 5 + 1.$$

Back-substitute:

$$1 = 6 - 5 = 6 - (17 - 2 \cdot 6) = 3 \cdot 6 - 17.$$

Therefore

$$3 \cdot 6 + (-1) \cdot 17 = 1.$$

Reducing modulo 17 gives

$$6 \cdot 3 \equiv 1 \pmod{17}.$$

Answer

$$[6]_{17}^{-1} = [3]_{17}.$$

Worked inverse example: $[23]_{73}$

Use the Euclidean algorithm:

$$73 = 3 \cdot 23 + 4,$$

$$23 = 5 \cdot 4 + 3,$$

$$4 = 1 \cdot 3 + 1.$$

Back-substitute:

$$1 = 4 - 3$$

$$= 4 - (23 - 5 \cdot 4) = 6 \cdot 4 - 23$$

$$= 6(73 - 3 \cdot 23) - 23 = 6 \cdot 73 - 19 \cdot 23.$$

Hence

$$-19 \cdot 23 \equiv 1 \pmod{73}.$$

So

$$[23]_{73}^{-1} = [-19]_{73} = [54]_{73}.$$

A class with no inverse

Consider $[6]_{15}$.

$$\gcd(6, 15) = 3 \neq 1.$$

Therefore $[6]_{15}$ is **not** invertible. Why not? If it had inverse $[k]_{15}$, then

$$6k \equiv 1 \pmod{15}.$$

But the left-hand side is always divisible by 3, while the right-hand side is not.

Conclusion

No inverse can exist unless a is relatively prime to n .

Cancellation when a factor is invertible

Cancellation law

If $\gcd(c, n) = 1$ and

$$ac \equiv bc \pmod{n},$$

then

$$a \equiv b \pmod{n}.$$

Reason.

Since $\gcd(c, n) = 1$, the class $[c]_n$ has inverse $[c]_n^{-1}$. Multiply both sides by that inverse. $\square$

Example

Since $\gcd(3, 8) = 1$, from

$$30 \equiv 6 \pmod{8}$$

we may divide by 3 to get

$$10 \equiv 2 \pmod{8}.$$

Why division can fail

Danger sign

If $\gcd(c, n) \neq 1$, then $[c]_n$ has no inverse, so “dividing by c ” is not legitimate.

Counterexample

$$30 \equiv 6 \pmod{8}$$

is true, but dividing by 2 would give

$$15 \equiv 3 \pmod{8},$$

which is false.

- ▶ 2 is not invertible modulo 8.
- ▶ In fact $[2]_8$ is a zero-divisor: $[2]_8[4]_8 = [0]_8$.

Every nonzero class is either invertible or a zero-divisor

Corollary

For $n > 1$, each nonzero element of $\mathbb{Z}_n$ is either

- ▶ invertible, or
- ▶ a zero-divisor,

but not both.

Reason:

- ▶ if $\gcd(a, n) = 1$, then $[a]_n$ is invertible;
- ▶ if $\gcd(a, n) = d > 1$, write $a = kd$ and $n = td$; then

$$[a]_n[t]_n = [0]_n, \quad [t]_n \neq [0]_n.$$

Example: invertible classes and zero-divisors in $\mathbb{Z}_{14}$

The nonzero classes are

$$[1]_{14}, [2]_{14}, \dots, [13]_{14}.$$

Invertible

Those relatively prime to 14:

$$[1]_{14}, [3]_{14}, [5]_{14}, [9]_{14}, [11]_{14}, [13]_{14}.$$

Zero-divisors

All the others:

$$[2]_{14}, [4]_{14}, [6]_{14}, [7]_{14}, [8]_{14}, [10]_{14}, [12]_{14}.$$

Prime modulus is special

Corollary

If p is prime, then every nonzero element of $\mathbb{Z}_p$ is invertible.

Why? For $1 \leq a \leq p - 1$, the only divisors of p are 1 and p , so $\gcd(a, p) = 1$.

Consequence

$\mathbb{Z}_p$ has no nonzero zero-divisors.

Examples:

- ▶ In $\mathbb{Z}_5$, all of $[1]_5, [2]_5, [3]_5, [4]_5$ are invertible.
- ▶ In $\mathbb{Z}_7$, every nonzero class has a unique inverse.

The set of invertible classes G_n

Definition

Let

$$G_n = \{[a]_n \in \mathbb{Z}_n : \gcd(a, n) = 1\}.$$

This is the set of all invertible classes modulo n .

Closure under multiplication

If $[a]_n, [b]_n \in G_n$, then

$$[a]_n[b]_n = [ab]_n \in G_n.$$

Reason: inverses multiply too,

$$([a]_n[b]_n)([a]_n^{-1}[b]_n^{-1}) = [1]_n.$$

Example: G_{20} and section summary

Example

The classes relatively prime to 20 are

$$G_{20} = \{[1]_{20}, [3]_{20}, [7]_{20}, [9]_{20}, [11]_{20}, [13]_{20}, [17]_{20}, [19]_{20}\}.$$

- ▶ $[3]_{20}^{-1} = [7]_{20}$ because $21 \equiv 1 \pmod{20}$.
- ▶ $[9]_{20}^{-1} = [9]_{20}$ because $81 \equiv 1 \pmod{20}$.
- ▶ $[11]_{20}^{-1} = [11]_{20}$ because $121 \equiv 1 \pmod{20}$.

Summary of Section 1.4

Congruence classes form a number system $\mathbb{Z}_n$ with addition and multiplication; invertibility is governed by $\gcd(a, n) = 1$.

Transition: from arithmetic to equations

Now that we can compute in $\mathbb{Z}_n$, we can solve equations there.

Main question

Given integers a, b, n with $n \geq 2$, when does

$$ax \equiv b \pmod{n}$$

have a solution? And how many solutions are there?

- ▶ Sometimes none.
- ▶ Sometimes exactly one class.
- ▶ Sometimes several classes modulo n .

What is a linear congruence?

A **linear congruence** is an equation of the form

$$ax \equiv b \pmod{n},$$

where x is the unknown integer. Examples:

$$6x \equiv 5 \pmod{17},$$

$$6x \equiv 5 \pmod{15},$$

$$6x \equiv 9 \pmod{15}.$$

These illustrate the three possibilities:

- ① no solution,
- ② one solution class,
- ③ several solution classes.

Main theorem for linear congruences

Theorem

Let $d = \gcd(a, n)$. The congruence

$$ax \equiv b \pmod{n}$$

has a solution if and only if $d \mid b$.

If $d \mid b$, then:

- ▶ there are exactly d solutions modulo n ;
- ▶ all solutions are congruent modulo n/d .

This theorem answers three questions at once

Existence, uniqueness, and the full list of solutions.

Why the divisibility condition is necessary

Suppose $x = c$ is a solution. Then

$$ac \equiv b \pmod{n},$$

so $n \mid (ac - b)$, hence for some $k \in \mathbb{Z}$,

$$ac - b = nk.$$

Rearrange:

$$b = ac - nk.$$

Now $d = \gcd(a, n)$ divides both terms on the right, so $d \mid b$.

Therefore

If a solution exists, then necessarily $\gcd(a, n) \mid b$.

Why the divisibility condition is sufficient

Now assume $d = \gcd(a, n)$ divides b . Write

$$b = de.$$

Because $d = \gcd(a, n)$, there exist integers k, t with

$$d = ak + nt.$$

Multiply by e :

$$b = ake + nte.$$

Hence

$$a(ke) \equiv b \pmod{n}.$$

Therefore

Whenever $d \mid b$, a solution exists: one solution is $x = ke$.

Why the reduced modulus n/d appears

If $x = c$ solves

$$ax \equiv b \pmod{n},$$

then

$$ac = b + nk.$$

Divide by d :

$$(a/d)c = b/d + (n/d)k.$$

So every solution also satisfies

$$(a/d)x \equiv b/d \pmod{n/d}.$$

Conversely, any solution of the reduced congruence gives a solution of the original one.

Key point

The real “shape” of the solution is one congruence class modulo n/d .

Why there are exactly d solutions modulo n

Assume c solves the reduced congruence modulo n/d . Then the corresponding solutions modulo n are

$$c, \quad c + \frac{n}{d}, \quad c + 2\frac{n}{d}, \quad \dots, \quad c + (d-1)\frac{n}{d}.$$

These are distinct modulo n and exhaust all possibilities.

Interpretation

One class modulo n/d splits into exactly d classes modulo n .

Algorithm for solving $ax \equiv b \pmod{n}$

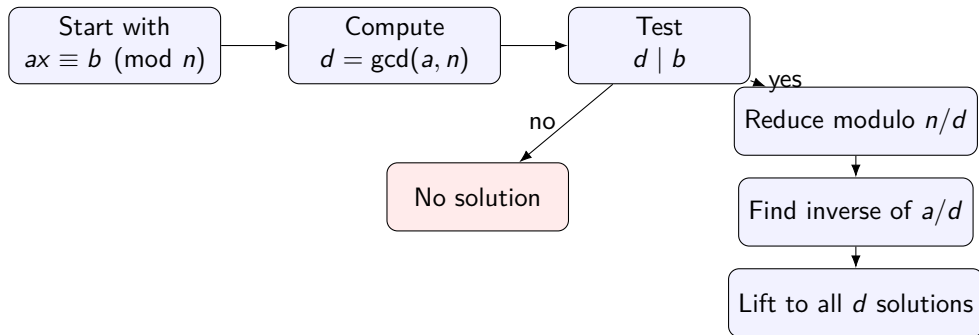
- 1 Compute $d = \gcd(a, n)$.
- 2 If $d \nmid b$, stop: no solution.
- 3 If $d \mid b$, divide through by d to get

$$(a/d)x \equiv b/d \pmod{n/d}.$$

- 4 Since $\gcd(a/d, n/d) = 1$, find the inverse of a/d modulo n/d .
- 5 Multiply by that inverse to get one solution c modulo n/d .
- 6 Lift to all d solutions modulo n :

$$c, c + n/d, \dots, c + (d - 1)n/d.$$

A compact problem-solving flowchart



Memory aid

gcd check $\rightarrow$ reduce $\rightarrow$ invert $\rightarrow$ lift

Example 1: a unique solution

Solve

$$6x \equiv 5 \pmod{17}.$$

- ▶ $\gcd(6, 17) = 1$, so there is one solution modulo 17.
- ▶ Need 6^{-1} modulo 17.
- ▶ Since $6 \cdot 3 = 18 \equiv 1 \pmod{17}$, we have $6^{-1} \equiv 3$.
- ▶ Multiply both sides by 3:

$$x \equiv 3 \cdot 5 = 15 \pmod{17}.$$

Answer

$$x \equiv 15 \pmod{17}.$$

Example 2: no solution

Solve

$$6x \equiv 5 \pmod{15}.$$

- ▶ $\gcd(6, 15) = 3$.
- ▶ But $3 \nmid 5$.

Conclusion

There is **no solution**.

Reason: the left side $6x$ is always divisible by 3, while the right side 5 is not.

Example 3: several solutions

Solve

$$6x \equiv 9 \pmod{15}.$$

- ▶ $d = \gcd(6, 15) = 3$, and $3 \mid 9$.
- ▶ Therefore there are exactly 3 solutions modulo 15.
- ▶ Divide through by 3:

$$2x \equiv 3 \pmod{5}.$$

- ▶ Since $2^{-1} \equiv 3 \pmod{5}$, we get

$$x \equiv 3 \cdot 3 = 9 \equiv 4 \pmod{5}.$$

So one reduced solution is $x \equiv 4 \pmod{5}$.

Lifting the reduced solution in Example 3

From the reduced congruence we found

$$x \equiv 4 \pmod{5}.$$

Since $d = 3$, the solutions modulo 15 are

$$4, \quad 4 + 5 = 9, \quad 4 + 10 = 14.$$

Hence the complete solution set is

$$[4]_{15}, \quad [9]_{15}, \quad [14]_{15}.$$

Check

$$6 \cdot 4 = 24 \equiv 9, \quad 6 \cdot 9 = 54 \equiv 9, \quad 6 \cdot 14 = 84 \equiv 9 \pmod{15}.$$

Example 4: a larger congruence

Solve

$$432x \equiv 12 \pmod{546}.$$

- ▶ $\gcd(432, 546) = 6$.
- ▶ Since $6 \mid 12$, there are exactly 6 solutions modulo 546.
- ▶ Divide by 6:

$$72x \equiv 2 \pmod{91}.$$

Now we need the inverse of 72 modulo 91.

Finishing Example 4

Find 72^{-1} modulo 91. By the Euclidean algorithm one gets

$$72^{-1} \equiv 67 \pmod{91}.$$

Therefore

$$x \equiv 67 \cdot 2 = 134 \equiv 43 \pmod{91}.$$

So the six solutions modulo 546 are

$$43, 134, 225, 316, 407, 498.$$

Answer

$$x \equiv 43, 134, 225, 316, 407, 498 \pmod{546}.$$

Common mistakes when solving linear congruences

- ❶ Forgetting to check $d = \gcd(a, n)$ first.
- ❷ Dividing by a even when a is not invertible modulo n .
- ❸ Finding only one solution when $d > 1$.
- ❹ Lifting by the wrong step size.
- ❺ Mixing up “mod n ” and “mod n/d ” after reduction.

Safe habit

Always write: **gcd check** $\rightarrow$ **reduce** $\rightarrow$ **invert** $\rightarrow$ **lift**.

Systems of congruences: a new problem

We may want an integer satisfying several remainder conditions at once:

$$x \equiv 3 \pmod{7}, \quad x \equiv 6 \pmod{25}.$$

Questions:

- ▶ Does a simultaneous solution exist?
- ▶ Is it unique modulo something?
- ▶ How do we construct it efficiently?

The answer is the **Chinese Remainder Theorem**.

Chinese Remainder Theorem (CRT)

Theorem

Suppose $m \geq 2$ and $n \geq 2$ are relatively prime, and $a, b \in \mathbb{Z}$. Then the system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

has a simultaneous solution. Moreover, the solution is unique modulo mn .

Idea

When the moduli are coprime, the remainder conditions do not interfere with each other.

Why CRT works: proof idea

If $\gcd(m, n) = 1$, then Bézout gives integers k, t such that

$$mk + nt = 1.$$

Now define

$$c = bmk + ant.$$

Then modulo m :

$$c \equiv ant \equiv a(nt) \equiv a \pmod{m},$$

because $nt \equiv 1 \pmod{m}$. Similarly, modulo n :

$$c \equiv bmk \equiv b \pmod{n}.$$

So c is a simultaneous solution.

Uniqueness in CRT

Suppose c and d are both solutions of

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}.$$

Then

$$c - d \equiv 0 \pmod{m} \quad \text{and} \quad c - d \equiv 0 \pmod{n}.$$

So both m and n divide $c - d$. Because $\gcd(m, n) = 1$, it follows that

$$mn \mid (c - d).$$

Conclusion

Any two simultaneous solutions are congruent modulo mn .

CRT example by the constructive formula

Solve

$$x \equiv 3 \pmod{7}, \quad x \equiv 6 \pmod{25}.$$

Since $\gcd(7, 25) = 1$, find integers with

$$7(-7) + 25(2) = 1.$$

Using the CRT construction,

$$x = 6 \cdot 7 \cdot (-7) + 3 \cdot 25 \cdot 2 = -144.$$

Reduce modulo 175:

$$-144 \equiv 31 \pmod{175}.$$

Answer

$$x \equiv 31 \pmod{175}.$$

The same CRT example by substitution

Start from

$$x \equiv 3 \pmod{7}.$$

Write

$$x = 3 + 7k.$$

Substitute into the second congruence:

$$3 + 7k \equiv 6 \pmod{25}.$$

So

$$7k \equiv 3 \pmod{25}.$$

Since $7^{-1} \equiv 18 \pmod{25}$,

$$k \equiv 18 \cdot 3 = 54 \equiv 4 \pmod{25}.$$

Thus

$$x = 3 + 7(4 + 25r) = 31 + 175r.$$

Again,

A three-congruence system: setup

Solve

$$x \equiv 2 \pmod{7}, \quad x \equiv 0 \pmod{9}, \quad 2x \equiv 6 \pmod{8}.$$

First simplify the third congruence:

$$2x \equiv 6 \pmod{8}.$$

Since $\gcd(2, 8) = 2$, divide by 2 to obtain

$$x \equiv 3 \pmod{4}.$$

So now solve

$$x \equiv 2 \pmod{7}, \quad x \equiv 0 \pmod{9}, \quad x \equiv 3 \pmod{4}.$$

A three-congruence system: solution

Combine the first two congruences first. A number divisible by 9 and congruent to 2 mod 7 is

$$9k \equiv 2 \pmod{7} \iff 2k \equiv 2 \pmod{7} \iff k \equiv 1 \pmod{7}.$$

So

$$x \equiv 9 \pmod{63}.$$

Now solve with the third condition:

$$x = 9 + 63t, \quad 9 + 63t \equiv 3 \pmod{4}.$$

Since $9 \equiv 1$ and $63 \equiv 3 \pmod{4}$,

$$1 + 3t \equiv 3 \pmod{4} \iff 3t \equiv 2 \pmod{4} \iff t \equiv 2 \pmod{4}.$$

Hence

$$x = 9 + 63(2 + 4r) = 135 + 252r.$$

Answer

A brief look beyond linear congruences

Not every congruence problem is linear.

► Quadratic example:

$$x^2 + 1 \equiv 0 \pmod{n}.$$

► Polynomial example:

$$x^3 - x^2 + x + 1 \equiv 0 \pmod{3}.$$

A useful strategy: if an integer equation has a solution, then it must have a solution modulo every n .

Contrapositive test

If a polynomial has *no* solution modulo some n , then it has no integer solution.

This is a powerful way to rule out integer roots.

In-class practice set

Try these during or after the lecture.

- 1 Compute all classes in $\mathbb{Z}_5$ and identify the inverses of the nonzero elements.
- 2 Decide whether $[10]_{21}$ is invertible.
- 3 Solve $15x \equiv 9 \pmod{24}$.
- 4 Solve $14x \equiv 8 \pmod{30}$.
- 5 Solve

$$x \equiv 1 \pmod{4}, \quad x \equiv 2 \pmod{5}.$$

Suggested classroom rhythm

5 minutes individual work, 5 minutes pair-check, 5 minutes whole-class discussion.

Full lecture summary

- ▶ Congruence modulo n groups integers by common remainder.
- ▶ The set $\mathbb{Z}_n$ has well-defined addition and multiplication.
- ▶ A class $[a]_n$ is invertible iff $\gcd(a, n) = 1$.
- ▶ Nonzero classes are either invertible or zero-divisors.
- ▶ The congruence $ax \equiv b \pmod{n}$ is solvable iff $\gcd(a, n) \mid b$.
- ▶ If solvable, the number of solutions modulo n is exactly $\gcd(a, n)$.
- ▶ For coprime moduli, the Chinese Remainder Theorem produces a unique solution modulo the product.

Exit ticket

Write down, in one sentence each:

- 1 when a class $[a]_n$ is invertible,
- 2 when $ax \equiv b \pmod{n}$ has a solution,
- 3 what extra hypothesis is needed for the Chinese Remainder Theorem.